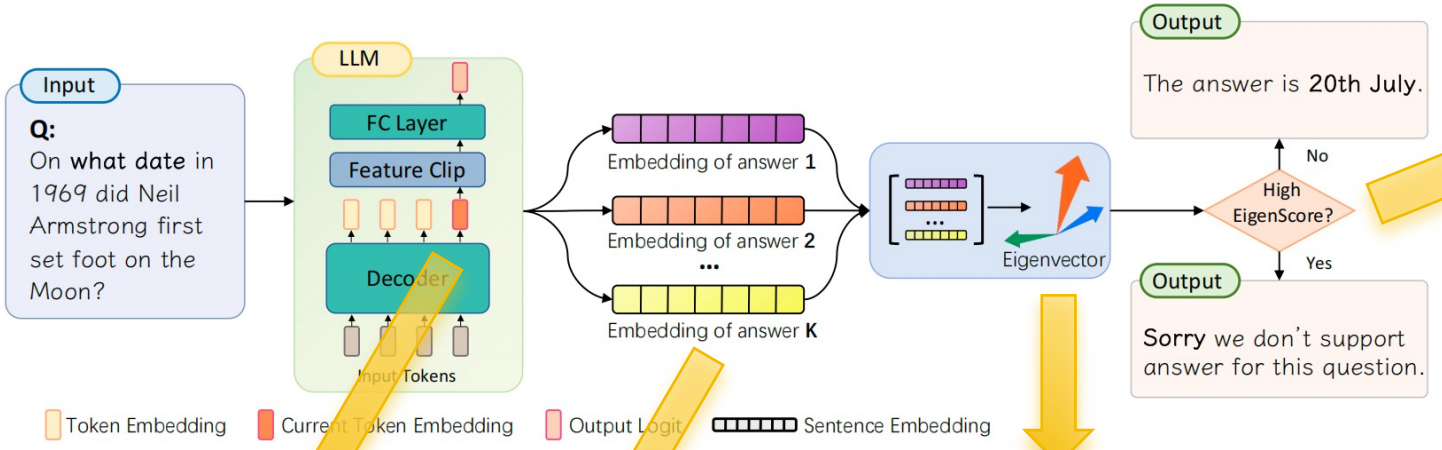
EigenScore

EigenScore[13]



Sentence Embedding Extraction

INSIDE takes the last-token **hidden state from a middle layer** as the sentence embedding z:

$$z = h_T$$

1. Covariance Matrix of K Sampled Embeddings

Given K generated answer embeddings:

$$Z = [z_1, z_2, \ldots, z_K] \in \mathbb{R}^{d \times K}$$

Covariance matrix:

$$\Sigma = Z^\top \cdot J_d \cdot Z$$

where

- $J_d = I_d - \frac{1}{d} 1_d 1_d^\top$ (centering matrix)

2. EigenScore Definition (LogDet of Covariance)

$$E(\mathcal{Y}|x,\theta) = \frac{1}{K} \log\det(\Sigma + \alpha I_K)$$

- αI_K: small regularization to make the covariance matrix full-rank

3. Equivalent Eigenvalue Expression

$$E(\mathcal{Y}|x,\theta) = \frac{1}{K} \sum_{i=1}^{K} \log(\lambda_i)$$

Conceptual Meaning

- **Large EigenScore** → embeddings spread out → *high uncertainty*
- **Small EigenScore** → embeddings clustered → *model is confident*

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EigenScore[13]

- **Task Type:** Open-ended generation

- **Introduction:** EigenScore constructs a representation-level uncertainty estimator by extracting last-token hidden states from a middle transformer layer and computing an EigenScore derived from the log-determinant of their covariance matrix across sampled generations. It quantifies decoding uncertainty by measuring divergence in the model's internal semantic representations, capturing how consistently the model encodes its own answers.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Consensus-based	Decoding	White-box	Post-hoc	Low	Ranking

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